

Euclid's Elements — Book I

Proposition 22

Formal Reconstruction — Technical Documentation

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Constructing a triangle from three given side lengths

1 Introduction

Euclid's Proposition I.22 is the classical existence theorem for a triangle with three prescribed side lengths.

Three nondegenerate segments are given. Euclid assumes that any two of them, taken together, are greater than the remaining one. Under these hypotheses he constructs a triangle whose three sides are congruent to the three given segments.

In the present reconstruction the given segments are denoted

$$Aa, \quad Bb, \quad Cc.$$

The three hypotheses are the strict triangle inequalities

$$Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb.$$

The formal conclusion is the existence of a point K such that

$$BK \cong Aa \quad \text{and} \quad bK \cong Cc.$$

Since Bb is already the third prescribed side, the triangle BbK has precisely the required side lengths.

Proposition I.22 is especially important for the formal development because its proof exposes two pieces of infrastructure which classical notation tends to hide:

1. the phrase “two segments taken together are greater than a third” must receive a synthetic, witness-based meaning;
2. the existence of the vertex K is an application of circle-circle continuity.

No numerical length function is introduced. Segment sums, strict inequalities, and the circle intersection are all expressed in the synthetic language of betweenness and congruence.

2 The Classical Configuration

Euclid's construction is best understood in the classical ray configuration used by Wilkins. Let the three prescribed lengths be represented by A , B , and C . On one ray choose consecutive segments

$$DF = A, \quad FG = B, \quad GH = C.$$

Draw the circle centered at F with radius A and the circle centered at G with radius C . Under the three strict triangle inequalities the two circles intersect properly in two points K and L . Either triangle FGK or triangle FGL then has side lengths A, B, C .

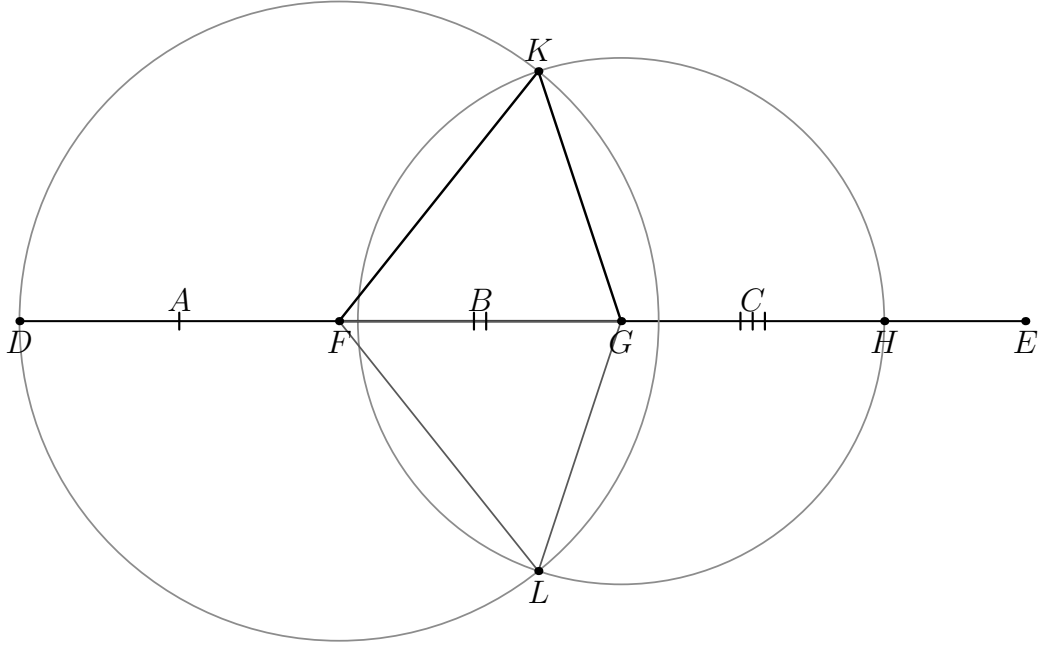


Figure 1: Wilkins-style classical configuration for Euclid I.22. The circles centered at F and G have radii represented by DF and GH and intersect in K and L . Either intersection produces a triangle with the three prescribed side lengths.

The strict triangle inequalities are not decorative side conditions. They are exactly what prevents tangency, separation, or containment. The six equality and reversed-inequality boundary states are shown in Figure 2.

3 Classical Proof

Euclid's proof is a direct ruler-and-compass construction. Let the three prescribed segments be represented by

$$Aa, \quad Bb, \quad Cc,$$

and assume

$$Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb.$$

Choose Bb as the base. Draw the circle centered at B with radius congruent to Aa and the circle centered at b with radius congruent to Cc . Let K be an intersection point. By construction,

$$BK \cong Aa, \quad bK \cong Cc,$$

so triangle BbK has the three prescribed side lengths.

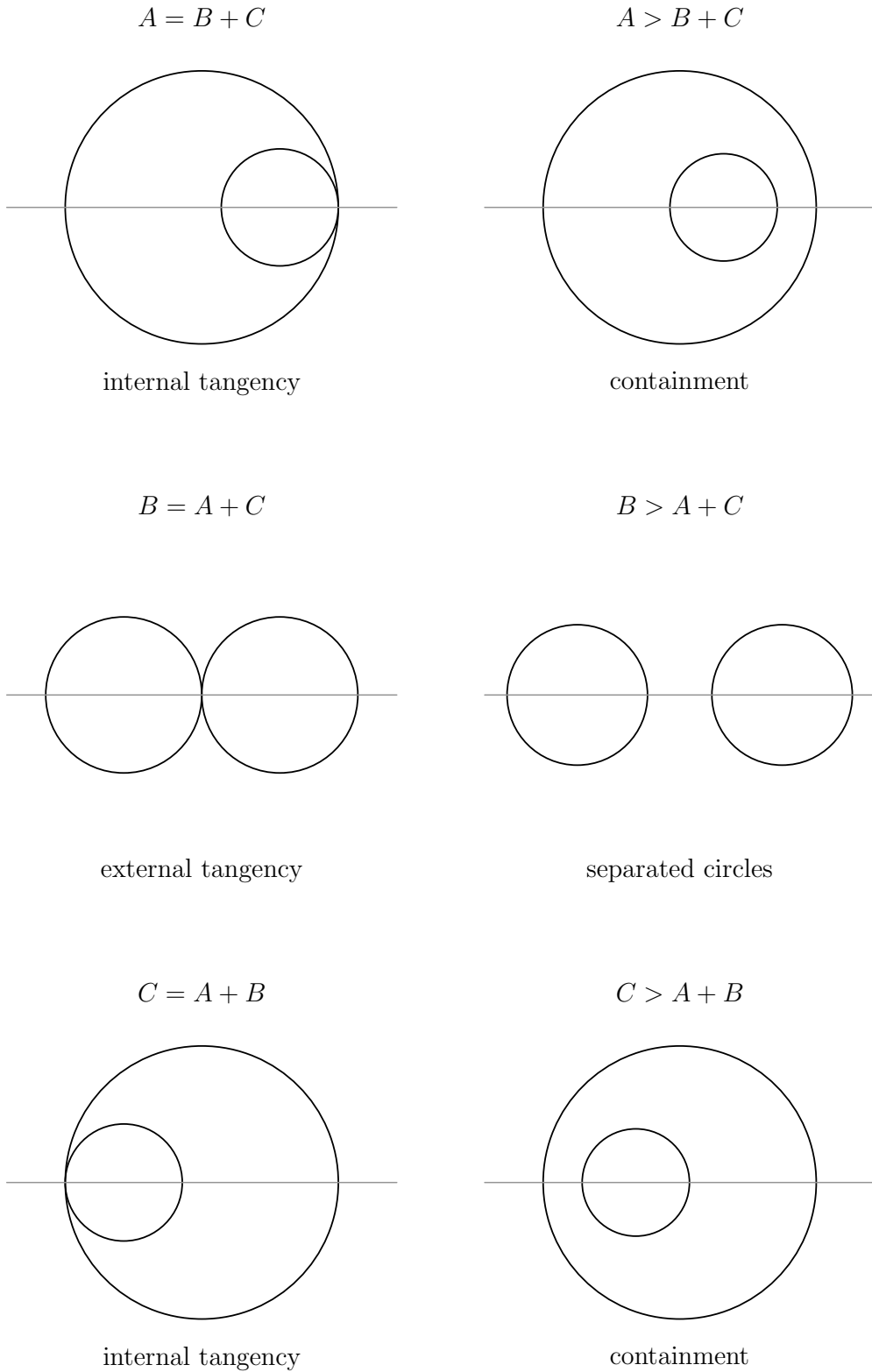


Figure 2: Classical boundary and failure configurations for the three triangle inequalities. Equality produces tangency; reversing an inequality destroys proper intersection. These six panels are classical admissibility states, not branches of the Lean proof.

The classical proof may be summarized as

$$\begin{array}{c}
Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb \\
\downarrow \\
\text{draw the two circles} \\
\downarrow \\
K \text{ is an intersection point} \\
\downarrow \\
BK \cong Aa, \quad bK \cong Cc \\
\downarrow \\
\triangle BbK.
\end{array}$$

The point at which the axiomatic reconstruction must say more is the existence of K . In the traditional diagram the circles are simply drawn as intersecting. The formal proof must derive the hypotheses of an intersection theorem from explicit segment-order data.

4 Statement

The synthetic statement used in the final Lean theorem is:

```

theorem euclid_proposition_22
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A a B b C c : Geo.Point)
  (hAa : A != a)
  (hBb : B != b)
  (hCc : C != c)
  (hAa_lt_BbCc :
    HilbertSegmentSumGreater Geo B b C c A a)
  (hBb_lt_AaCc :
    HilbertSegmentSumGreater Geo A a C c B b)
  (hCc_lt_AaBb :
    HilbertSegmentSumGreater Geo A a B b C c) :
  exists K : Geo.Point,
    Geo.Congruent B K A a /\
    Geo.Congruent b K C c

```

The three assumptions involving `HilbertSegmentSumGreater` express

$$Bb + Cc > Aa, \quad Aa + Cc > Bb, \quad Aa + Bb > Cc.$$

The conclusion gives the third vertex K of the required triangle:

$$BK \cong Aa, \quad bK \cong Cc,$$

while the base itself is the given segment Bb .

5 A Synthetic Predicate for “Two Together Greater”

Proposition I.21 showed that segment sums need not in general be reified as numerical objects. A concrete collinear segment can serve as a representative of a sum.

Proposition I.22 introduces a slightly different requirement. The three triangle inequalities are not merely intermediate steps in the proof: they are part of the public hypotheses of the theorem. Consequently it is useful to package the geometric witness explicitly.

The Book Zero predicate is

```
def HilbertSegmentSumGreater
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point) : Prop :=
  exists P : Geo.Point,
    Geo.Between A B P /\
    Geo.Congruent B P C D /\
    HilbertSegmentLess Geo E F A P
```

Geometrically, a witness P satisfies

$$A - B - P, \quad BP \cong CD.$$

Therefore AP represents the sum

$$AB + CD.$$

The final condition

$$EF < AP$$

says precisely

$$AB + CD > EF.$$

For example,

`HilbertSegmentSumGreater Geo B b C c A a`

means that there exists P with

$$B - b - P, \quad bP \cong Cc, \quad Aa < BP.$$

Thus BP is a synthetic representative of $Bb + Cc$.

The public theorem contains three such hidden witnesses. Their incidence/order content is recorded in Figure 3; the points are ordinary geometric points, not numerical sum objects.

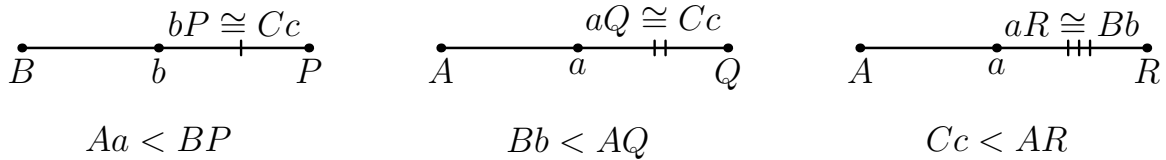


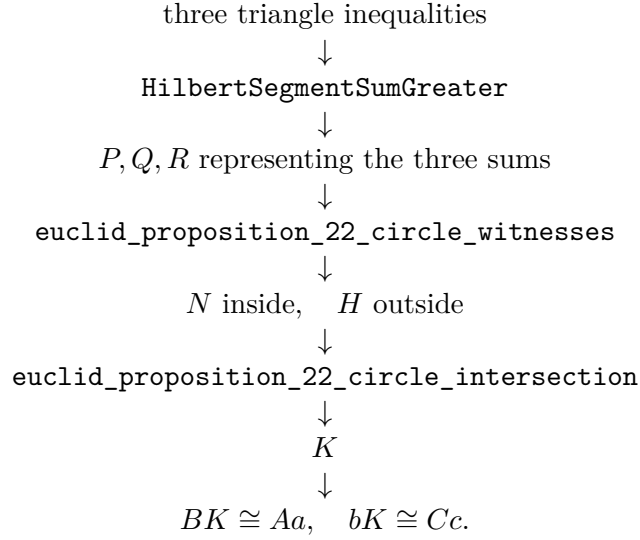
Figure 3: The three witnesses unpacked from the public hypotheses `HilbertSegmentSumGreater`: $B - b - P$ with $bP \cong Cc$ and $Aa < BP$; $A - a - Q$ with $aQ \cong Cc$ and $Bb < AQ$; and $A - a - R$ with $aR \cong Bb$ and $Cc < AR$. They are three independent realizations of the three synthetic sums.

This definition does not introduce an algebra of lengths. It merely packages an ordinary geometric construction together with the strict comparison already provided by `HilbertSegmentLess`.

6 Proof Architecture of the Reconstruction

The reconstruction is intentionally layered. The final theorem is short because the difficult circle geometry is isolated in reusable helpers.

The overall dependency is



The principal intermediate results are discussed below.

6.1 Transporting the outside inequality

The helper

`euclid_proposition_22_outside_helper`

handles the synthetic arithmetic behind the outside point.

Suppose BP represents

$$Bb + Cc,$$

and another collinear segment FH is assembled from pieces

$$FG \cong Bb, \quad GH \cong Cc.$$

Using `bookZero_sumOfParts`, the proof obtains

$$BP \cong FH.$$

Hence the triangle inequality

$$Aa < BP$$

can be transported to

$$Aa < FH$$

by `bookZero_30_lessThanCongruence`.

This is the formal replacement for the classical step “equal sums may be substituted in an inequality.”

6.2 Same-ray trichotomy

A point constructed on a ray can lie before, at, or beyond the reference point. The helper

`euclid_proposition_22_sameRay_trichotomy`

makes this order distinction explicit.

For points G, F, N on the same ray from G , it proves

$$G - N - F \quad \vee \quad N = F \quad \vee \quad G - F - N.$$

In the main I.22 construction one substitutes $G := b$ and $F := B$. The exact three geometric branches are therefore shown in Figure 4.

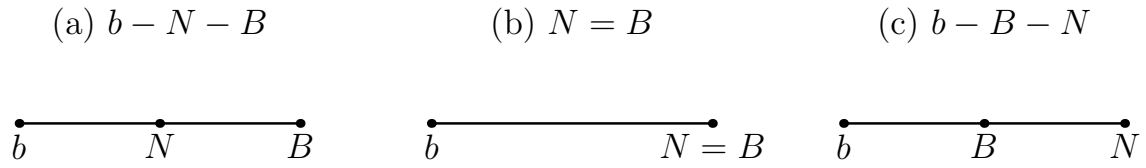


Figure 4: The same-ray trichotomy used for the inside witness N : $b - N - B$, $N = B$, or $b - B - N$. The first and third branches are handled by `euclid_proposition_22_inside_left` and `euclid_proposition_22_inside_right`; the middle branch is the center case.

The proof uses strict segment trichotomy together with `bookZero_51_lessThanBetween` and layoff uniqueness.

This case distinction is essential when the point N on the second circle is compared with the first circle.

6.3 Cancellation for strict segment inequalities

The helper

`euclid_proposition_22_lessThan_cancel_right`

is a synthetic cancellation principle.

If

$$A - B - E, \quad C - D - F,$$

the terminal parts satisfy

$$BE \cong DF,$$

and

$$AE < CF,$$

then

$$AB < CD.$$

Classically this is an immediate cancellation of equal addends. Formally it requires a proof from the developed strict-order and congruence theory.

This helper is one of the clearest examples in I.22 of Book Zero providing arithmetic-like behavior without numerical lengths.

6.4 The inside point

The proof constructs a point N on the ray from b through B such that

$$bN \cong Cc.$$

The two possible order configurations are handled by

```
euclid_proposition_22_inside_left
euclid_proposition_22_inside_right
euclid_proposition_22_inside_helper
```

and the same-ray trichotomy theorem.

The goal is not merely to locate N on the second circle. One must also prove that it is inside, or on the center of, the first circle:

$$N = B \quad \vee \quad BN < Aa.$$

The relevant triangle inequalities are used to derive this comparison in each possible collinear configuration.

6.5 Constructing explicit sum representatives

The helper

```
euclid_proposition_22_extend_with_segment
```

constructs the ordinary geometric representative of a sum.

Starting from a nondegenerate segment AB and another segment CD , it extends the ray through B to a point P satisfying

$$A - B - P, \quad BP \cong CD.$$

Hence

$$AP$$

represents

$$AB + CD.$$

This is the generic geometric operation of adjoining a congruent terminal segment. In the current public theorem, however, the witnesses P, Q, R are *not* constructed by this helper: they are unpacked directly from the three hypotheses of type `extttHilbertSegmentSumGreater`. On the actual main DAG, `exttt euclid_proposition_22_extend_with_segment` is called inside `exttt euclid_proposition_22_circle_witnesses` to construct the outside witness H with

$$B - b - H, \quad bH \cong Cc.$$

6.6 Circle witnesses

The theorem

`euclid_proposition_22_circle_witnesses`

is the main preparatory result.

From the three represented triangle inequalities it constructs points H and N on the circle centered at b with radius congruent to Cc , with the following properties:

$$bH \cong Cc, \quad Aa < BH,$$

and

$$bN \cong Cc, \quad N = B \vee BN < Aa.$$

Thus H is outside the circle centered at B with radius Aa , while N is inside it or coincides with its center.

This is precisely the configuration required by circle-circle continuity.

7 Circle-Circle Continuity

The central existence step is isolated in

`euclid_proposition_22_circle_intersection`

The geometric input and output of the continuity step are shown in Figure 5. Panel (a) depicts the strict-inside branch $BN < Aa$; the alternative center case $N = B$ has already been recorded in Figure 4. Panel (b) records the genuinely new intersection witness K .

The Hilbert circle-continuity theorem does not use the radius Aa directly as a numerical quantity. The proof first lays off a congruent segment BT :

$$BT \cong Aa.$$

The first circle is therefore represented by center B and radius segment BT .

The second circle has center b and radius bN , with

$$bN \cong Cc.$$

The already constructed points satisfy

$$bH \cong bN,$$

so H lies on the second circle, and

$$Aa < BH.$$

After transport through $BT \cong Aa$, this becomes

$$BT < BH,$$

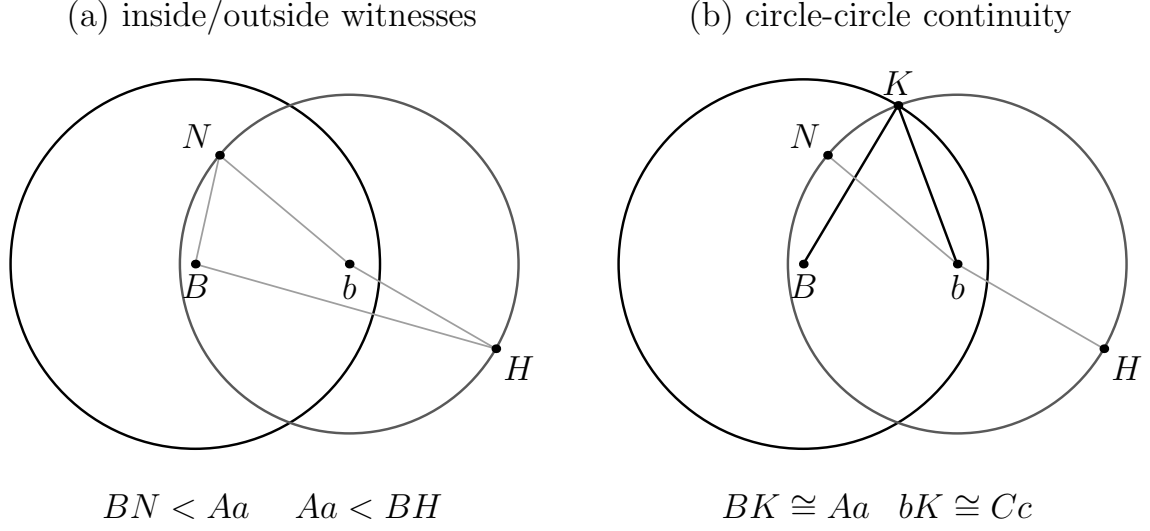


Figure 5: Formal circle-continuity chain. Both N and H lie on the circle centered at b with radius congruent to Cc . In the strict-inside branch, $BN < Aa$ while $Aa < BH$; circle-circle continuity then supplies a common point K , after which congruence transport yields $BK \cong Aa$ and $bK \cong Cc$.

showing that H is outside the first circle.

Similarly,

$$N = B \quad \vee \quad BN < Aa$$

is transported to

$$N = B \quad \vee \quad BN < BT,$$

showing that N is inside the first circle.

The theorem

`hilbert_circle_circle_intersection`

then yields a point K satisfying

$$BK \cong BT \quad \text{and} \quad bK \cong bN.$$

Finally congruence transitivity returns to the original given segments:

$$BK \cong Aa, \quad bK \cong Cc.$$

This is the required vertex.

8 The Core and the Public Theorem

Once the circle witnesses and the circle-intersection theorem are available, the remaining layers become very short.

The theorem

euclid_proposition_22_core

accepts explicit representatives P, Q, R of the three sums:

$$BP = Bb + Cc, \quad AQ = Aa + Cc, \quad AR = Aa + Bb,$$

together with

$$Aa < BP, \quad Bb < AQ, \quad Cc < AR.$$

It invokes the two successive helpers

euclid_proposition_22_circle_witnesses
euclid_proposition_22_circle_intersection

The intermediate wrapper

euclid_proposition_22_from_sum_witnesses

packages the same construction when all three sum witnesses are supplied explicitly.

The public theorem finally removes P, Q, R from its interface. It opens the three witnesses stored in `HilbertSegmentSumGreater` and passes them to the internal construction.

Thus the public API states exactly the geometric theorem rather than the implementation details of its proof.

9 Lean Formalization

The current production theorem exposes only the three prescribed segments, their nondegeneracy, the three witness-bearing triangle inequalities, and the required vertex:

```
theorem euclid_proposition_22
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A a B b C c : Geo.Point)
  (hAa : A != a)
  (hBb : B != b)
  (hCc : C != c)
  (hAa_lt_BbCc :
    HilbertSegmentSumGreater Geo B b C c A a)
  (hBb_lt_AaCc :
    HilbertSegmentSumGreater Geo A a C c B b)
  (hCc_lt_AaBb :
    HilbertSegmentSumGreater Geo A a B b C c) :
  Exists fun K : Geo.Point =>
    Geo.Congruent B K A a /\
    Geo.Congruent b K C c := by
  rcases hAa_lt_BbCc with
    <P, hBbP, hbP_Cc, hAa_BP>
  rcases hBb_lt_AaCc with
    <Q, hAaQ, haQ_Cc, hBb_AQ>
  rcases hCc_lt_AaBb with
```

```

    <R, hAaR, haR_Bb, hCc_AR>
exact
euclid_proposition_22_from_sum_witnesses
  Geo A a B b C c P Q R
  hAa hBb hCc
  hAa_BP hBb_AQ hCc_AR
  hBbP hbP_Cc
  hAaQ haQ_Cc
  hAaR haR_Bb

```

Thus the public endpoint is a wrapper over a substantial proposition-specific construction rather than over a single pre-existing interface theorem. The main internal chain is

$$\begin{array}{c}
 P, Q, R \text{ unpacked from the three sum hypotheses} \\
 \downarrow \\
 \text{euclid_proposition_22_circle_witnesses} \\
 \downarrow \\
 H, N \text{ with outside/inside data} \\
 \downarrow \\
 \text{euclid_proposition_22_circle_intersection} \\
 \downarrow \\
 K \\
 \downarrow \\
 \text{euclid_proposition_22.}
 \end{array}$$

The module also still contains two public declarations with historical test names,

```

euclid_proposition_22_outside_transport_test
euclid_proposition_22_inside_point_test

```

which isolate earlier infrastructure checks. They are not on the final dependency path from the public theorem to K .

No local `axiom` or `sorry` declaration occurs in the current `Proposition22.lean`. This audit did not run `lake build` or `#print axioms`; therefore it makes no new global axiom-audit claim beyond the source-level observation and the already documented use of circle-circle continuity.

10 Relationship with Earlier Propositions

10.1 Proposition I.20

I.20 proves the triangle inequality from an already existing triangle. I.22 uses the converse pattern as an existence hypothesis: three nondegenerate segments satisfying all three strict triangle inequalities are required to determine a constructible triangle. The logical directions are therefore different.

10.2 Proposition I.21

I.21 showed that segment sums can be handled synthetically by ordinary constructed collinear segments. I.22 promotes that technique to the public interface through the witness-bearing predi-

cate **HilbertSegmentSumGreater**. The resulting witnesses P, Q, R are geometric representatives of the three sums used as hypotheses.

10.3 Proposition I.23

In Euclid's historical dependency graph, I.23 uses I.22 essentially to build an auxiliary triangle before applying SSS. In the Hilbert reconstruction, I.23 does not theorem-depend on I.22 because Hilbert III.4 already supplies angle construction. I.22 therefore marks an early and useful separation between the historical Book I DAG and the DAG induced by the Hilbert foundation.

11 Hilbert and Book Zero Components Used

11.1 HilbertSegmentLess

The relation

HilbertSegmentLess

is the synthetic strict order on segments.

It supplies the comparisons needed to distinguish points lying inside and outside a circle and supports the transport and transitivity operations used throughout the proof.

11.2 bookZero_49_layoff

The theorem

bookZero_49_layoff

copies a given nondegenerate segment onto a chosen ray.

In I.22 it is used to represent the radius Aa by a segment based at the center B :

$$BT \cong Aa.$$

11.3 bookZero_sumOfParts

The theorem

bookZero_sumOfParts

recognizes congruent wholes assembled from congruent adjacent parts.

It is used when a segment representing $Bb + Cc$ must be identified with another segment built from a copy of Bb followed by a copy of Cc .

11.4 Congruence transport for inequalities

The theorems

`bookZero_30_lessThanCongruence`
`bookZero_32_lessThanCongruence2`

transport strict segment comparisons through congruent representatives.

They replace the informal algebraic operation of substituting equal lengths into inequalities.

11.5 `bookZero_31_trichotomy1`

This theorem supplies the equality case when neither of two nondegenerate segments is strictly shorter than the other.

Together with layoff uniqueness it provides the middle case

$$N = F$$

in the same-ray trichotomy used by I.22.

11.6 `bookZero_51_lessThanBetween`

The theorem

`bookZero_51_lessThanBetween`

converts strict segment comparison on a common ray into strict betweenness.

It is the bridge between the order relation on segment lengths and the incidence-order geometry needed to locate the constructed circle points.

11.7 Circle-circle intersection

The theorem

`hilbert_circle_circle_intersection`

is the decisive existence principle.

Given a point of one circle lying inside a second circle and another point of the same first circle lying outside the second, it produces a common point of the two circles.

Proposition I.22 is therefore not merely an application of elementary segment arithmetic. It is the point where that arithmetic is used to establish the hypotheses of a genuine continuity theorem.

12 Comparison with Euclid’s Original Organization

Euclid presents I.22 as an explicit ruler-and-compass construction. He places copies of the three given segments on a straight line and draws two circles whose intersection gives the required triangle.

The Hilbert reconstruction preserves the same geometric idea but changes its logical organization.

First, the base Bb is already available as one of the given segments, so there is no need to reproduce Euclid’s preliminary line solely to manufacture a base of the correct length.

Second, the three assumptions “two together greater than the remaining one” are exposed through **HilbertSegmentSumGreater**. Their witnesses give concrete segments representing the required sums.

Third, the existence of the intersection is not treated as visually obvious. Explicit points N and H are constructed on one circle, and they are proved to lie respectively inside and outside the other. Only then is circle-circle continuity invoked.

Thus the formal proof separates three issues which Euclid’s diagram compresses:

$$\text{segment arithmetic} \quad + \quad \text{order on rays} \quad + \quad \text{continuity}.$$

The resulting proof is longer, but the additional length identifies the exact logical content required for the construction.

13 What Proposition I.22 Adds to Book Zero

I.21 suggested that explicit segment-sum objects were unnecessary: inside a proof, sums could be represented directly by constructed segments.

I.22 refines that conclusion rather than contradicting it.

The new predicate

$$\text{HilbertSegmentSumGreater}$$

does not turn segment sums into algebraic objects. It is a witness-bearing proposition:

$$AB + CD > EF$$

means that a point exists which geometrically realizes $AB + CD$, and that the resulting whole segment is strictly greater than EF .

This distinction is important.

The project still has no numerical length map and no binary addition operation on lengths. What has been added is a convenient *interface proposition* for a classical phrase that occurs as a hypothesis of Euclid’s theorem.

Proposition I.22 also shows that the Book Zero layer is now strong enough to prepare the exact inside/outside data required by a continuity theorem. The layer is therefore doing more than abbreviating elementary congruence arguments: it is connecting synthetic segment arithmetic with topological existence principles.

14 Dependency Summary

The public assumptions are

$$\begin{aligned} Aa \neq 0, \quad Bb \neq 0, \quad Cc \neq 0, \\ Aa < Bb + Cc, \\ Bb < Aa + Cc, \\ Cc < Aa + Bb. \end{aligned}$$

In Lean the three inequalities are encoded by

```
HilbertSegmentSumGreater Geo B b C c A a,
HilbertSegmentSumGreater Geo A a C c B b,
HilbertSegmentSumGreater Geo A a B b C c.
```

The complete proof architecture is

```
triangle inequalities
↓
P, Q, R sum witnesses
↓
segment transport + same-ray order
↓
N = B ∨ BN < Aa    Aa < BH
↓
bN ≅ Cc,    bH ≅ Cc
↓
hilbert_circle_circle_intersection
↓
BK ≅ Aa,    bK ≅ Cc
↓
euclid_proposition_22.
```

No new geometric axiom is introduced in Proposition I.22. The proof uses the already available circle-circle continuity theorem together with the developed Hilbert and Book Zero segment theory.

15 Formalization Notes

Several features of I.22 are easy to miss in an informal proof.

First, a circle radius is represented by an ordinary segment, not by a real number. Even when the intended radius is Aa , the circle-continuity theorem is applied using a constructed representative BT with

$$BT \cong Aa.$$

Second, “inside a circle” is expressed synthetically by a strict segment comparison such as

$$BN < BT,$$

with the center case $N = B$ handled separately.

Third, “outside a circle” is likewise a strict segment comparison:

$$BT < BH.$$

Fourth, the fact that N lies on the ray from b through B is not enough to determine its order relative to B . The proof requires an explicit three-way analysis of the possible positions.

Fifth, cancellation of equal terminal pieces from strict inequalities is not automatic. It is isolated as a theorem and proved from the existing segment-order machinery.

Finally, the public theorem is deliberately cleaner than the internal construction. The implementation works with many auxiliary points, but the theorem exposes only the three original segments, their three triangle inequalities, and the required vertex.

16 Diagrammatic Audit

The preserved `Book1R.pdf` contained two figures for I.22, both classical. Figure 22.1 recorded the Wilkins-style proper-intersection construction, while Figure 22.2 recorded the six tangency/separation/containment boundary states. Both are retained because they represent genuinely different geometric configurations.

The audit adds three formal figures required by the current Lean DAG:

- Figure 3 records the hidden witnesses P, Q, R carried by the three public sum-greater hypotheses;
- Figure 4 records the exact three-way order split for N on the ray from b through B ;
- Figure 5 records the inside/outside circle-continuity state and the subsequent appearance of the intersection witness K .

No separate figure is added for the radius representative T , for congruence/inequality transport, for `bookZero_sumOfParts`, for the cancellation helper, or for the wrapper layers. Those steps change the formal representation or logical information without creating a new proposition-specific incidence/order configuration. In particular, T is an important formal witness for the radius segment based at B , but drawing it as an additional geometric state would duplicate the same first circle rather than record new geometry.

The final diagram set therefore consists of five figures: two classical and three formal. The six panels of the classical boundary map are not Lean branches, while the three panels of the same-ray figure are an actual exhaustive formal case split.

17 Conclusion

Proposition I.22 is a major structural milestone in the reconstruction of Book I.

Its classical content is elementary: three admissible lengths determine a constructible triangle. Formalizing that statement reveals, however, a precise interaction between three layers of synthetic geometry.

The first layer is segment arithmetic. Betweenness, congruence, strict segment comparison, addition of adjacent parts, cancellation, and transport are used to express the three triangle inequalities and to manipulate their witnesses.

The second layer is ray and order geometry. Constructed points must be located explicitly on rays, and their possible orders must be distinguished.

The third layer is continuity. Once one point of the second circle is proved inside the first and another outside it, `hilbert_circle_circle_intersection` supplies the required common point.

The final theorem therefore has a particularly clear conceptual form: the triangle inequalities are exactly the hypotheses needed to build the inside/outside witnesses for two circles.

No coordinates, numerical lengths, or angle measures enter the proof. The reconstruction remains entirely synthetic, and the complete development through Euclid I.22 compiles successfully.